CALCULUS · UNIT 3: DERIVATIVES

Sprint Coach Analytics

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students coach a 100-meter sprinter. Timing-gate data gives average and estimated instantaneous velocities, a position model is differentiated to find velocity and acceleration, and other runners' models require the product, quotient, and chain rules.

Tangent lines estimate split times between gates.

Standards Addressed

- AP CHA-2 Derivative as instantaneous rate; definition
- AP FUN-3 Derivative rules: power, product, quotient, chain
- AP CHA-3 Rates of change in motion
- AP CHA-3.F Tangent-line approximation (preview)

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a slow-motion sprint video. Part 1 data and rates.
Day 2	Part 2 model derivatives.
Day 3	Part 3 rules practice and the report.

Materials

- Student handout and Timing Gate Data
- Graphing calculators
- Optional: stopwatch and tape measure to collect class data

How the Three Versions Differ

Level	What changes
CP2	Average velocity, symmetric estimates, limit definition for $0.5t^2$, power rule, and a tangent line.
CP1	Data-based acceleration, exponential model derivatives, product and quotient rules, and a tangent estimate.
Honors	Fitting parameters, predicted race times, chain and quotient rules with stride models, jerk, and runner comparison.

Watch For (Common Misconceptions)

- Confusing average and instantaneous velocity.
- Forgetting the chain-rule factor for e^(−0.4t).
- Interpreting negative acceleration as moving backward.

Extension Ideas

- Collect sprint data from your gym class.
- Compare with world-record splits.
- Model a 200-meter race.

Answer Key – CP2

Part 1: Average Velocity

#	Task	Answer
1	Find the average velocity from t = 0 to t = 11 s.	$\frac{85.3-0}{11} \approx 7.75$ m/s
2	Find the average velocity from t = 1 to 2 s and from t = 9 to 10 s. What do you notice?	4.4 m/s and 9.8 m/s; the sprinter speeds up.
3	Estimate the instantaneous velocity at t = 5 s using t = 4 and t = 6.	$\frac{37.3-20.0}{2} \approx 8.65$ m/s
4	When does the sprinter seem to reach top speed? Use the data.	Around t = 9 to 11 s (about 9.8 m/s)

Part 2: Power Rule

#	Task	Answer
5	Use the limit definition to find s′(t) for $s(t) = 0.5t^2$.	$\frac{0.5(t+h)^2 - 0.5t^2}{h} = t + 0.5h \to t$
6	Find the velocity at t = 4 and t = 8.	4 m/s and 8 m/s
7	Use the power rule to find the derivative of $s(t) = 0.2t^3 + t$.	$s'(t) = 0.6t^2 + 1$
8	Find the acceleration a(t) = v′(t) for $s(t) = 0.5t^2$.	$a(t) = 1$ m/s^2

Part 3: Tangent Lines

#	Task	Answer
9	Write the tangent line at t = 6.	s(6) = 18, s′(6) = 6 → y − 18 = 6(t − 6)
10	Use the tangent line to estimate s(6.2). Compare with the actual value.	18 + 6(0.2) = 19.2; actual 19.22
11	Write two coaching tips based on your velocity results.	Answers vary; e.g., work on the start (slow early velocity).

Reflection

#	Prompt	Look for
R1	How is instantaneous velocity different from average velocity?	It is the rate at one instant (the limit of average velocities over shrinking intervals).
R2	What does a positive acceleration mean for the sprinter?	The velocity is increasing; the runner is speeding up.

Answer Key – CP1

Part 1: Data to Derivatives

#	Task	Answer
1	Estimate s′(2), s′(5), and s′(10) using symmetric differences.	$\frac{12.5-1.8}{2} \approx 5.35$, ≈ 8.65, $\frac{85.3-65.7}{2} = 9.8$ m/s
2	Estimate the acceleration at t = 5 using velocity estimates at t = 4 and t = 6 (s′(4) ≈ 7.95, s′(6) ≈ 9.05).	$\frac{9.05-7.95}{2} \approx 0.55$ m/s²
3	Is the sprinter's velocity increasing at a decreasing rate? Explain with the data.	Yes; the velocity grows but by smaller amounts each second (concave down velocity).
4	Sketch the velocity graph from your estimates.	Increasing curve leveling off near 10 m/s

Part 2: Model Derivatives

#	Task	Answer
5	Find v(t) = s′(t).	v(t) = 10 − 10e^(−0.4t)
6	Find a(t) = v′(t).	a(t) = 4e^(−0.4t)
7	Find v(5) and a(5). Compare with your data estimates.	v(5) ≈ 8.65 m/s; a(5) ≈ 0.54 m/s²
8	What value does v(t) approach? Interpret.	10 m/s; the sprinter's top speed

Part 3: Rules Practice

#	Task	Answer
9	Runner B: s(t) = t²(12 − t) ÷ 4 for 0 ≤ t ≤ 8. Use the product rule to find v(t). When is v greatest?	$v = \frac{24t - 3t^2}{4}$; v′ = 0 at t = 4 → v(4) = 12 m/s
10	Runner C: s(t) = $\frac{12t^2}{t+2}$. Use the quotient rule to find v(t) and v(4).	$v = \frac{12t^2+48t}{(t+2)^2}$; v(4) = $\frac{384}{36} \approx 10.7$ m/s
11	Write the tangent line to s(t) = 10t − 25 + 25e^(−0.4t) at t = 8 and estimate the position at 8.5 s.	s(8) ≈ 56.0, v(8) ≈ 9.59 → s(8.5) ≈ 60.8 m

Reflection

#	Prompt	Look for
R1	Why does the acceleration approach 0 as the race goes on?	The sprinter approaches top speed, so the velocity stops changing.
R2	How did the data estimates compare with the model's derivatives?	They were close, which supports the model.

Answer Key – Honors

Part 1: Model Fitting

#	Task	Answer
1	Show that the given model s(t) = 10t − 25 + 25e^(−0.4t) has this form. What are v_{max} and k?	v_{max} = 10, k = 0.4: 10t − 25(1 − e^(−0.4t))
2	Find v(t) and show v(0) = 0.	v = 10 − 10e^(−0.4t); v(0) = 0
3	Predict the 100-m time using the model (solve s(t) = 100 numerically).	t ≈ 12.5 s
4	The coach improves the start so k = 0.5 (same top speed). Find the new 100-m time.	10t − 20(1 − e^(−0.5t)) = 100 → t ≈ 12.0 s

Part 2: Rules & Rates

#	Task	Answer
5	The sprinter's stride length is L(v) = 0.18v + 0.6 m. Find $\frac{dL}{dt}$ at t = 3 using the chain rule.	$\frac{dL}{dt}$ = 0.18 v′(t) = 0.18(4e^(−1.2)) ≈ 0.217 m/s
6	Stride rate is r = $\frac{v}{L(v)}$ strides per second. Find $\frac{dr}{dv}$ using the quotient rule.	$\frac{dr}{dv} = \frac{0.6}{(0.18v+0.6)^2}$
7	Is stride rate increasing as velocity increases? Explain.	Yes; $\frac{dr}{dv} > 0$ for all v
8	Find the jerk (derivative of acceleration) at t = 2 and interpret its sign.	j = −1.6e^(−0.8) ≈ −0.72 m/s³; acceleration is decreasing

Part 3: Comparing Runners

#	Task	Answer
9	Find when Runner B's acceleration is 0 and her velocity then.	a = $\frac{24-6t}{4}$ = 0 at t = 4; v = 12 m/s
10	Compare positions at t = 4 and t = 8. Who leads?	A: s(4) ≈ 20.0, s(8) ≈ 56.0; B: s(4) = 32, s(8) = 64. B leads at both.
11	Write the coaching report recommending one training focus for each runner, citing derivatives.	Answers vary.

Reflection

#	Prompt	Look for
R1	What does the parameter k control in the sprint model?	How quickly the runner approaches top speed (the start).
R2	Why is the chain rule needed for stride length over time?	Stride depends on velocity, which depends on time.